CHENNAI INSTITUTE OF TECHNOLOGY, CHENNAI

Department of Computer Science and Engineering | Project-Based Learning (PBL) — Machine Learning | A.Y. 2026–2027

SUGARSENSE — EARLY-STAGE DIABETES RISK PREDICTION

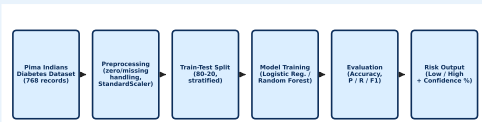
ABSTRACT

SugarSense predicts a patient's early-stage diabetes risk (Low / High) from eight routine clinical measurements — such as Glucose, BMI, Blood Pressure and Age — using Logistic Regression and a Random Forest trained on the Pima Indians Diabetes Dataset, and reports a risk label with a confidence percentage.

PROBLEM STATEMENT

- Early-stage diabetes is largely asymptomatic and is often missed until complications appear.
- Routine measurements (glucose, BMI, blood pressure, age) are rarely combined into one risk estimate at the point of care.
- Zero/missing values and class imbalance (65:35) make reliable prediction difficult.

SYSTEM ARCHITECTURE & METHODOLOGY



Data flows from the Pima Indians Diabetes Dataset through preprocessing (zero/missing-value handling, StandardScaler) and an 80-20 stratified split into Logistic Regression and Random Forest models, which are evaluated on accuracy, precision, recall and F1 and output as a Low/High risk label with a confidence percentage.

Tech stack: Python 3, pandas, scikit-learn, NumPy (Streamlit demo planned).

Dataset: 768 patient records — 8 clinical features, 2 classes (non-diabetic ≈ 65%, diabetic ≈ 35%). 80-20 stratified split: 614 train / 154 test.

OBJECTIVES

- Preprocess the Pima Indians Diabetes Dataset (768 records, 8 clinical features).
- Build and refine Logistic Regression and Random Forest classifiers.
- Evaluate using accuracy, precision, recall and F1-score.
- Document weekly progress and mentor feedback through the PBL cycle.

RELATED WORK

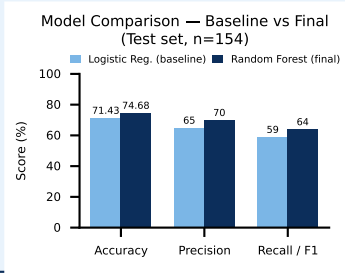
Prior work on the Pima dataset applies Logistic Regression, Decision Trees, SVM and Random Forest: Logistic Regression (~77%) is an interpretable baseline, while ensembles (~80%) capture non-linear feature interactions. Zero values are treated as missing, and class imbalance is handled with class weights or SMOTE.

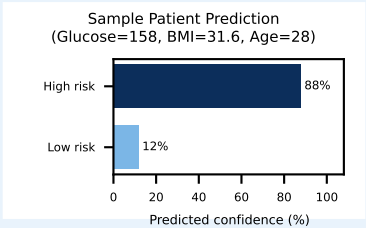
MAJOR RESULTS

74.68%

Test Accuracy · Random Forest (Final)

Final model, test set n=154. Precision 0.70 · Recall / F1 0.64





DISCUSSION | KEY CODE

The Random Forest outperformed the Logistic Regression baseline on every metric (accuracy 71.43% → 74.68%, recall/F1 0.59 → 0.64) — consistent with ensembles capturing non-linear feature interactions and noisy values.

Strong on clear high-glucose, high-BMI cases (88% confidence), weaker near the decision boundary. Results are pre-tuning; diabetic-class recall is the main area to improve — a screening aid, not a diagnosis.

KEY CODE · Python / scikit-learn

```
# Scale features (fit on training set only)
scaler = StandardScaler()
X_train_scaled = scaler.fit_transform(X_train)
# Train the Random Forest
rf_model = RandomForestClassifier(random_state=42)
rf_model.fit(X_train_scaled, y_train)
# Evaluate on the held-out test set
y_pred = rf_model.predict(scaler.transform(X_test))
print(accuracy_score(y_test, y_pred))  # 0.7468
```

CONCLUSION

SugarSense shows that a Random Forest trained on routine clinical measurements can provide a useful first-pass diabetes risk screen— 74.68% test accuracy, ahead of the Logistic Regression baseline (71.43%). Data quality (missing values, class imbalance) is the main limit; hyperparameter tuning and a Streamlit demo are the next steps.

FUTURE SCOPE | TEAM

- Scale to a larger, more diverse diabetes dataset.
- Tune with GridSearchCV; handle imbalance (SMOTE / class weights).
- Add 5-fold cross-validation for more rigorous evaluation.
- Deploy as a Streamlit app for real-time risk prediction.

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